DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2012.

CLASSICAL ALGEBRA

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Examine the convergence of the series $\frac{1}{1^k}+\frac{x}{3^k}+\frac{x^2}{5^k}+\cdots+\frac{x^{n-1}}{(2n-1)^k}+\cdots$ (10)

(b) Prove that $\frac{1}{1\cdot 3}+\frac{2}{3\cdot 5}+\frac{3}{5\cdot 7}+\frac{4}{7\cdot 9}+\cdots$ is divergent. (10)

2. (a) Test for convergence of the series $\sum_{n=0}^{\infty}\frac{n^3+1}{2^n+1}$. (10)

(b) Examine the convergence of the series $\sum\frac{(n+1)(n+2)\cdots(n+n)}{n^n}$. (10)

3. (a) Find the co-efficient of $x^n$ in the expansion of $\frac{1+2z-3x^2}{e^x}$. (10)

(b) Show that

$$\log\sqrt{2}=1+\left(\frac{1}{2}+\frac{1}{3}\right)\frac{1}{4}+\left(\frac{1}{4}+\frac{1}{5}\right)\frac{1}{4^2}+\left(\frac{1}{6}+\frac{1}{7}\right)\frac{1}{4^3}+\cdots$$

(10)

4. (a) Sum the series

$$1+\frac{1+3}{2!}+\frac{1+3+3^2}{3!}+\frac{1+3+3^2+3^3}{4!}+\cdots \text{ to } \infty \qquad (10)$$

(b) Sum the series to infinity

$$\frac{15}{16}+\frac{15\cdot 21}{16\cdot 24}+\frac{15\cdot 21\cdot 27}{16\cdot 25\cdot 32}+\cdots \qquad (10)$$

5. (a) If $\alpha, \beta, \gamma, \delta$ be the roots of the biquadratic equation $x^4+px^3+qx^2+rx+s=0$, find

(i) $\sum \alpha^2$

(ii) $\sum \alpha^2\beta\gamma$

(iii) $\sum \alpha^2\beta^2$

(iv) $\alpha^3\beta$

(v) $\sum \alpha^4$. (10)

(b) Find $\frac{1}{\alpha^5}+\frac{1}{\beta^5}+\frac{1}{\gamma^5}$ where $\alpha, \beta, \gamma$ are the roots of the equation $x^3+2x^2-3x-1=0$. (10)

6. (a) Solve the equation $x^4+4x^3+5x^2+2x-6=0$. (10)

(b) Solve the equation $81x^3-18x^2-36x+8=0$. (10)

7. (a) Prove that $\frac{1}{2\sqrt{n+1}}<\frac{1}{2}\cdot\frac{3}{4}\cdot\frac{5}{6}\cdots\frac{2n-1}{2n}<\frac{1}{\sqrt{2n+1}}$. (10)

(b) Prove that

$$\begin{vmatrix} a^2 & (s-a)^2 & (s-a)^2 \\ (s-b)^2 & b^2 & (s-b)^2 \\ (s-c)^2 & (s-c)^2 & c^2 \end{vmatrix} = 2s^3(s-a)(s-b)(s-c)$$

where $2s=a+b+c$. (10)

8. (a) Solve the following simultaneous equations

$$x + 2y + 5z = 23$$

$$3x + y + 4z = 26$$

$$6x + y + 7z = 47 .$$ (10)

(b) Diagonolise the matrix $\begin{bmatrix} 2 & -2 & 3 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$. (10)

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2012.

CALCULUS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

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$(5 \times 20 = 100)$

1. (a) If $y = e^{m \sin^{-1}(x)}$, prove that $(1 - x^2)y_2 - xy_1 = m^2 y$.

   (b) If $u = \sqrt{x^2 + y^2 + z^2}$ show that

   $$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial u}{\partial z}\right)^2 = 1.$$

2. (a) Find the equation of normal on the curve :

   $x = a\cos^3\theta$, $y = b\sin^3\theta$ at the point $\theta = \frac{\pi}{4}$.

   (b) Prove that the equation of evolute of the parabola is $27ay^2 = 4(x - 2a)^3$.

3. (a) Evaluate :

   (i) $\int \frac{dx}{a^2 - x^2}$.

   (ii) $\int \frac{dx}{\sqrt{x^2 - a^2}}$.

(b) Prove that :

(i) $\int_0^a f(x)dx = \int_0^a f(a-x)\,dx$ .

(ii) $\int_a^b f(x)dx = -\int_b^a f(x)\,dx$ .

4. (a) Prove that $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}}\,dx$ .

(b) Find the reduction formula for $\int \tan^n x dx$. Hence evaluate $\int \tan^4 x dx$.

5. (a) Solve : $\left(x^2 + y^2\right) dx - 2xy\,dy = 0$ .

(b) Solve : $y^2 + xyp - x^2p^2 = 0$ .

6. (a) Solve : $\left(4D^2 + 16D - 9\right)y = e^{x/2}$ .

(b) Solve : $\left(D^2 + a^2\right)y = x\cos ax$ .

7. (a) Find : $L\left[\sinh 2t + \sin^2 t\right]$.

(b) Find : $L^{-1}\left[\frac{2(S+1)}{(S^2+2S+5)^2}\right]$.

8. (a) Solve : $(mz - ny)p + (nx - lz)q = ly - mx$ .

(b) Solve $z = px + qy + c\sqrt{1 + p^2 + q^2}$ .

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DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2012.

ANALYTICAL GEOMETRY AND VECTOR CALCULUS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Show that two of the straight lines $ax^3 + bx^2y + cxy^2 + dy^3 = 0$ will be perpendiculas to each other if $a^2 + d^2 + bd + ac = 0$.

   (b) Show that the circles $x^2 + y^2 - 4x + 6y + 8 = 0$ and $x^2 + y^2 - 10x - 6y + 14 = 0$ touch and find the co-ordinates of the point of contact.

2. (a) If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents two straight lines equidistant from the origin. Show that $f^4 - g^4 = c(bf^2 - ag^2)$.

   (b) Prove that the tangents at the extremities of any focal choral of a conic intersect on the corresponding directrix.

3. (a) The direction cosines of two lines satisfy $l + m + n = 0$, $2lm + 2mn - nl = 0$. Find them.

   (b) Find the equation of the plane parallel to the plane $x + y + 2z = 1$ and through the point (1, 1, –1).

4. (a) Find the equation of the plane passing through the points (3, –8, 2) and (–2, –1, 1) and perpendicular to the plane $x+y+z+1=0$.

(b) Find the image of the point (1, 3, 4) in the plane $2x-y+z+3=0$.

5. (a) Prove that $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-2}{3}=\frac{y-3}{4}=\frac{z-4}{5}$ are coplanar and find the equation of the plane containing them.

(b) Find the shortest distance between the lines

$$\frac{x-5}{3}=\frac{y-6}{-4}=z-9;$$

$$2x-2y+z-3=0=2x-y+2x-9.$$

6. (a) Find the equation of the sphere that passes through the centre of the sphere $x^2+y^2+z^2+4x-6y+2z-1=0$ and the circle $x^2+y^2+z^2+2x-4z-11=0$, $x+2y+2z-3=0$.

(b) Find the equation of tangent plane at a point $x_1, y_1, z_1$ to the sphere $x^2+y^2+z^2+2ux+2vy+2wz+d=0$.

7. (a) Show that $\overline{F}=\left(6xy+z^3\right)\overline{i}+\left(3x^2-z\right)\overline{j}+\left(3xz^2-y\right)\overline{k}$ is irrotational and find $\phi$ such that $\overline{F}=\nabla\phi$. (8)

(b) Prove that $\operatorname{div}\left(\overline{A}\times\overline{B}\right)=\overline{B}\cdot\operatorname{curl}\,\overline{A}-\overline{A}\cdot\operatorname{curl}\overline{B}$. (6)

(c) Prove that $\operatorname{grad} r^n=nr^{n-2}\overline{r}$. (6)

8. (a) Use Gauss divergence theorem find $\iint_S \overline{F} \cdot \hat{n}\, ds$ where $\overline{F} = x^3\overline{i} + y^3\overline{j} + 2^3\overline{h}$ and $S$ is the surface of the sphere $x^2 + y^2 + z^2 = a^2$.

(b) Verify Stoke's theorem for $\overline{F} = \left(x^2 - y^2\right)\overline{i} + 2xy\,\overline{j}$ in the rectangular resion in the $X \circ Y$ plane bounded by the lines $x = 0,\ x = a,\ y = 0$ and $y = b$.

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**DE–3622** **23**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2012.

MECHANICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) The magnitude of resultant of two given forces of magnitudes $P$ and $Q$, is R. If $Q$ is doubled, then $R$ is doubled. If $Q$ is reversed, then $R$ is doubled. Show that $P:Q:R=\sqrt{2}:\sqrt{3}:\sqrt{2}$. (10)

(b) Three forces act along the bisectors of angles $A,B,C$ of a triangle $ABC$. Show that, if the forces are in equilibrium, then their magnitudes are in the ratio $\cos\frac{A}{2}:\cos\frac{B}{2}:\cos\frac{C}{2}$. (10)

2. (a) Let $R$ be the resultant of the forces $F_1$ and $F_2$ acting at a point $O$. Any transversal meets the lines of action of $F_1,F_2,R$ respectively at $A_1,A_2,B$. Show that $\frac{F_1}{OA_1}+\frac{F_2}{OA_2}=\frac{R}{OB}$ where $F_1=|F_1|$, $F_2=|F_2|$ and $R=|R|$. (10)

(b) $A$ and $B$ are two fixed points on a horizontal line at a distance '$C$' apart. Two light strings $AC$ and $BC$ of lengths '$b$' and '$a$' respectively support a mass at $c$. Show that the tensions of the strings are in the ratio $b(a^2 + c^2 - b^2) : a(b^2 + c^2 - a^2)$. (10)

3. (a) To find the resultant of two parallel forces acting on a rigid body. (10)

(b) Three forces acting along the sides of a triangle in the same order are equivalent to a couple. Show that they are independent to a couple. Show that they are proportional to the sides of the triangle. (10)

4. (a) Prove that the sum of the moments of two intersecting or parallel forces about any point is equal to the moment of the resultant of the forces about the same point. (10)

(b) A solid cone of height $h$ and semi vertical angle $\alpha$ is placed with its base against a smooth vertical wall and is supported by a string attached to its vertex to a point in the wall. Show that the greatest possible length of the string is $h\sqrt{1 + (16/9)\tan^2 \alpha}$. (10)

5. (a) To show that the path of a projectile is a parabola. (12)

(b) A particle is projected at an angle $\alpha$ with a velocity $u$ and it strikes up an inclined plane of inclination $\beta$ at right angles to the plane. Prove that $\cot \beta = 2 \tan(\alpha - \beta)$. (8)

6. (a) A mass $m$ after falling freely through a distance '$a$' begins to raise a mass $M$ greater than itself and connected with it by means of an inextensible string passes over a fixed pulley. Show that $M$ will have returned to its original position at the end of time $\frac{2m}{M-m}\sqrt{\frac{2a}{g}}$. (10)

(b) A smooth sphere of mass impinges obliquely on a smooth sphere of Mass M which is at rest. Show that if $m = eM$, the directions of motion after impact are at right angles. (e is the coefficient of restitution). (10)

7. (a) To obtain the differential equation of a central orbit in polar co ordinates. (10)

(b) A particle describes an elliptic orbit under a central force towards one focus $S$. If $v_1$ is the speed at the end $B$ of the minor axis and $v_2, v_3$ the speeds at $A, A'$ of the major axis. Show that $v_1^2 = v_2 v_3$. (10)

8. (a) To obtain period of oscillation of a simple pendulum. (10)

(b) A particle moves in a plane with an acceleration which is always directed to a fixed point 0 in the plane. Find the differential equation of its path. (10)

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DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2012.

ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) If $(M,d)$ is a metric space, prove that $d_1$ defined by $d_1(x, y) = \min\{1, d(x, y)\}$ is a metric on $M$.

(b) Prove that in any metric space the intersection of a finite number of open sets is open.

(c) Let $(M, d)$ be a metric space. Define $\rho(x, y) = \dfrac{d(x, y)}{1 + d(x, y)}$ prove that $d$ and $\rho$ are equivalent metrics on $M$. (6 + 7 + 7 = 20)

2. (a) Prove that $f : (M_1, d_1) \to (M_2, d_2)$ is continuous iff $f^{-1}(G)$ is open in $M_1$ whenever $G$ is open in $M_2$.

(b) Let $(M,d)$ be a metric space and $A \subseteq M$. Prove that $Int(A)$ is the union of all open sets contained in $A$.

(c) Prove that $Z$ has no limit point. (7 + 7 + 6 = 20)

3. (a) Let $(M,d)$ be a metric space and $A \subseteq M$. Prove that $x \in \overline{A}$ iff there is a sequence $(x_n)$ in $A$ such that $(x_n) \to x$.

(b) State and prove the Baire's category theorem. (10 + 10 = 20)

4. (a) Prove that a subspace of $R$ is connected iff it is an interval.

(b) State and prove the intermediate value theorem. (10 + 10 = 20)

5. (a) Prove that a closed subspace of a compact metric space is compact.

(b) In a metric space $M$, prove that the following are equivalent :

(i) $M$ is compact

(ii) Any infinite subset of $M$ has a limit point

(iii) $M$ is sequentially compact

(iv) $M$ is totally bounded and complete. (6 + 14 = 20)

6. (a) Prove that any closed interval $[a, b]$ is a compact subset of $R$.

(b) Prove that a metric space $M$ is compact iff any family of closed sets with finite intersection property has non-empty intersection. (10 + 10 = 20)

7. (a) If $f_n : [0,1] \to R$ is defined by $f_n(x) = x^n$ prove that the sequence $(f_n)$ converges pointwise to a function $f : [0,1] \to R$ given by $f(x) = \begin{cases} 0 \ if \ 0 \le x < 1 \\ 1 \ if \ x = 1 \end{cases}$.

(b) State and prove Dini's theorem. (8 + 12 = 20)

8. (a) Prove that the uniform limit of a sequence of continuous function is continuous.

(b) State and prove contraction mapping theorem. (10 + 10 = 20)

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**DE–3624** **25**

DISTANCE EDUCATION

B.Sc.(Mathematics) DEGREE EXAMINATION, MAY 2012.

PROBABILITY AND STATISTICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

(5 × 20 = 100)

1. (a) Let $S$ be a sample space. Let $A \subseteq S$. Then prove that

(i) $P(A) = 1 - P(\overline{A})$ where $\overline{A}$ is the complement of $A$

(ii) $P(\phi) = 0$. (10)

(b) A bag contains 5 red and 3 black balls and a second bag 4 red and 5 black balls. One of the bag is selected at random and a draw of 2 balls is made from it. What is the probability that one of them is red and the other is black? (10)

2. (a) A random variable $X$ has the following probability distribution :

| | | | | |
|---|---|---|---|---|
| $x$ : | –1 | 0 | 1 | 2 |
| $P(x)$ : | 1/3 | 1/6 | 1/6 | 1/3 |

Find the expectation of $X$. (10)

(b) Obtain the

(i) mean

(ii) median and

(iii) mode for the following distribution

$$f(x)=\begin{cases}6\left(x-x^2\right) if\ 0<x<1 \\ 0 \qquad\qquad \text{elsewhere}\end{cases}$$ (10)

3. (a) (i) Define skewness and kurtosis. (4)

(ii) Calculate the Karl pearson's coefficient of skewness.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Size : | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| Frequency : | 3 | 6 | 9 | 13 | 8 | 5 | 4 |

(b) Find the correlation coefficient from the following data : (10)

| | | | | | | |
|---|---|---|---|---|---|---|
| $x$ : | 10 | 12 | 18 | 24 | 23 | 27 |
| $y$ : | 13 | 18 | 12 | 25 | 30 | 10 |

4. (a) From the following data of marks obtained by 10 students in physics (P) and chemistry (C). Calculate the rank correlation coefficient. (10)

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| P : | 35 | 56 | 50 | 65 | 44 | 38 | 44 | 50 | 15 | 26 |
| C : | 50 | 35 | 70 | 25 | 35 | 58 | 75 | 60 | 55 | 35 |

(b) Out of the two lines of regression given by $x+2y-5=0$ and $2x+3y-8=0$. Which one is the regression line of $x$ on $y$? (10)

5. (a) In a Binomial distribution, the mean is 4 and the variance is 8/3. Find the mode of the distribution. (10)

(b) If $X$ is a Poisson random variate such that $P(X=1)=0.3$ and $P(X=2)=0.2$ . Find $P(X=0)$. (10)

6. (a) Prove that the mean for normal distribution is $\sigma\sqrt{2/\pi}$. (10)

(b) A random variable $X$ has the probability function $P(x) = \frac{1}{2^x}, x = 1, 2, 3, \ldots$ Find its moment generating function and variance. (10)

7. (a) A group of 10 rats fed on a diet A and another group of 8 rats fed on a different diet B recorded the following increase in weight in gms.

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Diet A : | 5 | 6 | 8 | 1 | 12 | 4 | 3 | 9 | 6 | 10 |
| Diet B : | 2 | 3 | 6 | 8 | 1 | 10 | 2 | 8 | – | – |

Test whether diet A is superior to diet B. (10)

(b) The theory predicts that the proportion of an object available in four groups A, B, C, D should be 9:3:3:1. In an experiment among 1600 items of this object the numbers in the four groups were 882, 313, 287 and 118. Use $\chi^2$ test to verify whether the experimental results supports the theory. (10)

8. (a) The following is the statistics showing the lives in hours of four batches of electric bulbs sold in different shops. Perform an analysis of variance and state your conclusion. (10)

| Batches | $S_1$ | $S_2$ | $S_3$ | $S_4$ | $S_5$ | $S_6$ | $S_7$ | $S_8$ |
|---|---|---|---|---|---|---|---|---|
| A | 1600 | 1610 | 1650 | 1680 | 1700 | 1720 | 1800 | – |
| B | 1580 | 1640 | 1640 | 1700 | 1750 | – | – | – |
| C | 1460 | 1550 | 1600 | 1620 | 1640 | 1660 | 1740 | 1820 |
| D | 1510 | 1520 | 1530 | 1570 | 1600 | 1680 | – | – |

(b) Construct Index numbers of prices by applying Fisher's index method and Laspeyre's index method. (10)

| Commodities | Base year | | Current year | |
|---|---|---|---|---|
| | Price | Qty | Price | Qty |
| A | 2 | 8 | 4 | 6 |
| B | 5 | 10 | 6 | 5 |
| C | 4 | 14 | 5 | 10 |
| D | 2 | 19 | 2 | 13 |

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2012.

ALGEBRA

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) If $\rho$ and $\sigma$ are equivalence relations defined on a set $S$, prove that $\rho \wedge \sigma$ is an equivalence relation. (5)

(b) Show that $f : R \to (0,1)$ defined by $f(x) = \frac{1}{2}\left[1 + \frac{x}{1+|x|}\right]$ is a bijection. (10)

(c) Let $f : A \to B$, $g : B \to C$ be bijections. Then $g \circ f : A \to C$ is also a bijection. (5)

2. (a) If $A$, $B$ and $C$ are three sets then

(i) $A\Delta B = (A \cup B) - (A \cap B)$

(ii) $A\Delta(B\Delta C) = (A\Delta B)\Delta C$. (10)

(b) If $A$, $B$ and $C$ are any three sets prove that

(i) $(A \cup B) \times C = (A \times C) \cup (B \times C)$

(ii) $(A - B) \times C = (A \times C) - (B \times C)$. (10)

3. (a) Let $A$ and $B$ be two subgroups of a group $G$. Then $AB$ is a subgroup of $G$ iff $AB = BA$. (10)

(b) Let $G$ be a group and $H$ be a subgroup of $G$. Then

(i) $a \in H \Rightarrow aH = H$

(ii) $aH = bH \Rightarrow a^{-1}b \in H$

(iii) $a \in bH \Rightarrow a^{-1} \in Hb^{-1}$

(iv) $a \in bH \Rightarrow aH = bH$. (10)

4. (a) State and prove fundamental theorem of homomorphism. (8)

(b) Let $G$ be any group. Show that $f : G \to G$ given by $f(x) = x^{-1}$ is an isomorphism $\Leftrightarrow G$ is abelian. (6)

(c) $H_a$ is the set of all elements in $G$ which commute with '$a$'. Show that $H_a$ is a subgroup of $G$. (6)

5. (a) Let $R$ be an Euclidean domain. Then any two elements $a, b \in R$ have a g.c.d. and it is of the form $ax + by$, where $x, y \in R$. (10)

(b) State and prove division algorithm. (10)

6. (a) Find the g.c.d. of $16 + 7i$ and $10 - 5i$ in the ring $R$ of Gaussian integers. (5)

(b) A non-empty subset $S$ of a ring $R$ is a subring iff $a, b \in S \Rightarrow a - b \in S$ and $ab \in S$. (5)

(c) Prove that the set of all matrices of the form $\begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix}$ where $a \in R$ is a ring under matrix addition and multiplication. (10)

7. (a) Let $V$ be a finite dimensional inner product space. Let $W$ be a subspace of $V$. Then $V$ is the direct sum of $W$ and $W^{\perp}$. (10)

(b) Let $V$ be a vector space over $F$ and $W$ a subspace of $V$. Let $\frac{V}{W} = \left\{ \frac{W+v}{v \in V} \right\}$. Then $\frac{V}{W}$ is a vector space over $F$ under the following operations :

(i) $(W + v_1) + (W + v_2) = W + v_1 + v_2$.

(ii) $\alpha(W + v_1) = W + \alpha v_1$. (10)

8. (a) Let $V$ and $W$ be vector spaces over a field $F$ and $T : V \to W$ be an epimorphism. Then

(i) Ker $T = V_1$ is a subspace of $V$ and

(ii) $\frac{V}{V_1} \cong W$. (10)

(b) Let $V$ be a finite dimensional vector space over a field $F$. Let $A$ and $B$ be subspaces of $V$. Then $\dim(A + B) = \dim A + \dim(A \cap B)$. (10)

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2012.

OPERATIONS RESEARCH

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Solve graphically the LPP :

Maximize $Z = 3x_1 + 2x_2$

Subject to : $5x_1 + x_2 \geq 10$

$x_1 + x_2 \geq 6$

$x_1 + 4x_2 \geq 12$

$x_1, x_2 \geq 0.$

(b) Explain Simplex method to find an optimal basic feasible solution to LPP.

2. (a) A farmer has 100 acres of land. He cultivates tomato, brinjal and green chilly in this land. He can sell all vegetables in the market and earn a profit of Rs. 2 per kg on tomatoes and brinjal and Rs. 3 per kg on green chillies. The average yield per acre in 2000 kgs of tomatoes, 3000 kgs of brinjals and 500 kgs of green chilly. Fertilizers are available at Rs. 5 per kg and the amount required per acre is 100 kgs each for tomatoes, brinjals and 50 kgs for green chillies. Labour required for sowing, cultivating and harvesting per acre is 5 man-days for tomatoes, 4 man days for brinjal and 6 man-days for green chillies. A total of 400 man-days of labour are available at Rs. 60 per man-day. Formulate this problem as an LPP to maximize the farmer's total profit.

(b) Using Simplex method, solve the following LPP :

Minimize $Z = x_1 - 3x_2 + 2x_3$

Subject to :
$$3x_1 - x_2 + 2x_3 \le 7$$
$$-2x_1 + 4x_2 \le 12$$
$$-4x_1 + 3x_2 + 8x_3 \le 10$$
$$x_1, x_2, x_3 \ge 0.$$

3. (a) Use the method of penalty to solve the following LPP :

Maximize $Z = 2x_1 + 3x_3$

Subject to :
$$x_1 + x_2 + 2x_3 \le 5$$
$$2x_1 + 3x_2 + 4x_3 = 12$$
$$x_1, x_2, x_3 \ge 0.$$

(b) Solve the following LPP using Two-phase Simplex method :

Minimize $Z = 60x_1 + 80x_2$

Subject to :
$$20x_1 + 30x_2 \ge 900$$
$$40x_1 + 30x_2 \ge 1200$$
$$x_1, x_2 \ge 0.$$

4. (a) Explain Two-phase Simplex method algorithm.

(b) Use Charnes penalty method to solve the LPP :

Maximize $Z = x_1 + 2x_2 + 3x_3 - x_4$

Subject to :
$$x_1 + 2x_2 + 3x_3 = 15$$
$$2x_1 + x_2 + 5x_3 = 20$$
$$x_1 + 2x_2 + x_3 + x_4 = 10$$
$$x_1, x_2, x_3, x_4 \ge 0.$$

5. (a) Explain North-West Corner method to find initial basic feasible solution to LPP.

(b) Using Row Minima method find a basic feasible solution to the following transportation problem :

| | $w_1$ | $w_2$ | $w_3$ | $a_i$ |
|---|---|---|---|---|
| $F_1$ | 8 | 10 | 12 | 900 |
| $F_2$ | 12 | 13 | 12 | 1000 |
| $F_3$ | 14 | 10 | 11 | 1200 |
| $b_j$ | 1200 | 1000 | 900 | 3100 |

6. Solve the transportation problem with transportation costs in rupees, requirements and availability as given below :

| | $D_1$ | $D_2$ | $D_3$ | $D_4$ | $a_i$ |
|---|---|---|---|---|---|
| $S_1$ | 10 | 15 | 12 | 12 | 200 |
| $S_2$ | 8 | 10 | 11 | 9 | 150 |
| $S_3$ | 11 | 12 | 13 | 10 | 120 |
| $b_j$ | 140 | 120 | 80 | 220 | 470 |
| | | | | | 560 |

7. (a) Determine which of the following two-person zero-sum are strictly determinable and fair. Give optimum strategies for each player in the case of strictly determinable games :

(i)

$$\begin{array}{cc} & \text{Player } B \\ \text{Player } A & \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix} \end{array}$$

(ii)

$$\begin{array}{cc} & B \\ A & \begin{bmatrix} 0 & 2 \\ -1 & 4 \end{bmatrix} \end{array}$$

(b) (i) Define zero-sum game.

(ii) Define payoff matrix.

8. (a) Solve the following $2 \times 2$ game graphically.

$$\begin{array}{cc} & \text{Player } B \\ & \begin{array}{cccc} B_1 & B_2 & B_3 & B_4 \end{array} \\ \text{Player } A \quad \begin{array}{c} A_1 \\ A_2 \end{array} & \begin{bmatrix} 2 & 1 & 0 & -2 \\ 1 & 0 & 3 & 2 \end{bmatrix} \end{array}$$

(b) Let $(a_{ij})$ be the $m \times n$ payoff matrix for a two-person zero-sum game. If $\underline{V}$ denotes the maximin value and $\overline{V}$ the minimax value of the game, then $\overline{V} \geq \underline{V}$. That is

$$\min_{1 \leq j \leq n}\left[\max_{1 \leq i \leq m}\{a_{ij}\}\right] \geq \max_{1 \leq i \leq m}\left[\min_{1 \leq j \leq n}\{a_{ij}\}\right].$$

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DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2012.

NUMERICAL METHODS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) (i) Explain Newton-Raphson method.

(ii) Find the real root of $x^3 - 3x + 1 = 0$ lying between 1 and 2 upto three decimal places using this method. (15)

(b) Explain Bisection method. (5)

2. Solve the equation

$x + y + z = 9$

$2x - 3y + 4z = 13$

$3x + 4y + 5z = 40$

(a) Gauss-Jordan method and (10)

(b) Gauss elimination method. (10)

3. (a) State and prove Newton's forward interpolation formula. (10)

(b) Using forward interpolation to find $f(79)$ from the following data : (10)

| | | | | |
|---|---|---|---|---|
| $x$ : | 75 | 80 | 85 | 90 |
| $f(x)$ : | 246 | 202 | 118 | 40 |

4. (a) Explain Stirling's formula. (8)

(b) Using Stirling's formula to find $y(25)$ from the following data : (12)

| | | | | |
|---|---|---|---|---|
| $x$ : | 20 | 24 | 28 | 32 |
| $y$ : | 2854 | 3162 | 3544 | 3992 |

5. Evaluate $\int_0^x \sin^2 x \, dx$ using

(a) Simpson's 1/3 rule and (10)

(b) Simpson's 3/8 rule. (10)

6. Evaluate $\int_0^1 \frac{dx}{1+x}$ using

(a) Trapezoidal rule and (10)

(b) Weddle's rule. (10)

7. Using Runge-Kutta method find $y(0.1)$, $y(0.2)$ and $y(0.3)$ given $\frac{dy}{dx} = 1 + xy$, $y(0) = 2$. (20)

8. Find an approximate solution of the initial value problem $y' = 1 + y^2$, $y(0) = 0$ by Picards method and compare with the exact solution. (20)

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DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2012.

COMPLEX ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) If $\frac{\partial^2}{\partial x \partial y} = \frac{\partial^2}{\partial y \partial x}$, prove that $\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = 4\frac{\partial^2}{\partial z \partial \bar{z}}$. (7)

(b) Prove that $f(z) = \sin x \cos hy + i \cos x \sin hy$ is differentiable at every point. (7)

(c) Prove that the real and imaginary parts of an analytic function are harmonic functions. (6)

2. (a) If $f(z) = u(x, y) + iv(x, y)$ is an analytic function and $u(x, y) = \frac{\sin 2x}{\cos h2y + \cos 2x}$, find $f(z)$. (12)

(b) If $u(x, y)$ is a harmonic function in a region $D$, prove that $f(z) = \frac{\partial u}{\partial x} - i\frac{\partial u}{\partial y}$ is analytic in $D$. (8)

3. (a) Find the bilinear transformation which maps $-1, 0, 1$ of the $z$-plane onto $-1, -i, 1$ of the $w-$plane. Show that under this transformation the upper half of the z-plane maps onto the interior of the unit circle $|w| = 1$. (14)

(b) Show that by means of inversion $w = \frac{1}{z}$, the circle given by $|z-3| = 5$ is mapped onto the circle $\left|w + \frac{3}{16}\right| = \frac{5}{16}$. (6)

4. (a) Prove that any bilinear transformation can be expressed as a product of translation, rotation, magnification or contraction and inversion. (12)

(b) Find the bilinear transformation which maps the points $z_1 = 0, z_2 = -i, z_3 = -1$ into $w_1 = i, w_2 = 1$, $w_3 = 0$ respectively. (8)

5. (a) Prove that $\int_c \frac{dz}{(z-a)^n} = \begin{cases} 0 & if \quad n \neq 1 \\ 2\pi i & if \quad n = 1 \end{cases}$ where $c$ is the circle with centre a and radius r and $n \in Z$. (10)

(b) State and prove Liouville's Theorem. (10)

6. (a) State and prove Taylor's Theorem. (15)

(b) Using higher derivative formula, evaluate $\int_c \frac{e^{2z}}{(z+1)^4} dz$ where c is the circle $|z| = 2$. (5)

7. (a) Prove that the function $f(z)$ comes arbitrarily close to any complex number c in every neighbourhood of an essential singularity. (10)

(b) State and prove Rouche's Theorem. (10)

8. (a) Evaluate $\int_0^{2\pi} \frac{d\theta}{13 + 5\sin\theta}$. (12)

(b) Prove that $\int_c \frac{e^{2z}}{(z+1)^3} dz = \frac{4\pi i}{e^2}$ where c is $|z| = \frac{3}{2}$. (8)

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2012.

DISCRETE MATHEMATICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Construct a truth table for the following compound propositions :

   (i) $(p \vee q) \to p \wedge q$

   (ii) $(p \to q) \to (q \to p)$

   (iii) $(q \to \neg p) \leftrightarrow (p \leftrightarrow q)$. (10)

   (b) Explain : Negation, Conjunction, Disjunction, Conditional Statement and Bi-conditional statement. (10)

2. (a) Indicate which ones are tautologies or contradictions :

   (i) $((P \to (\neg P)) \to \neg P)$

   (ii) $(P \to (P \vee Q))$

   (iii) $(P \to (Q \to P))$

   (iv) $((\neg Q \wedge P) \wedge Q)$. (10)

   (b) Obtain the principal disjunctive normal form for $(p \to (q \wedge r)) \wedge (\neg p \to (\neg q \wedge \neg r))$. (10)

3. (a) Demonstrate that $r$ is a valid inference from the premises $p \to q$, $q \to r$, $P$. (10)

(b) Test the validity of the following argument :

If I study, then I will not fail in mathematics. If I do not play basket ball, then I will study. But I failed in Mathematics. Therefore I must have played Basket Ball. (10)

4. (a) Show that, $(\forall x)(H(x) \to M(x)) \wedge H(S) \Rightarrow M(S)$. (10)

(b) Show that $(\rceil x) M(x)$ follows logically from the premises $(\forall x)(H(x) \to M(x))$ and $((\rceil x) H(x))$. (10)

5. (a) (i) Define directed graph and undirected graph with an example. (4)

(ii) Show that the sum of degree of all the vertices in a graph $G$ is even. (6)

(b) (i) Define a cycle graph with an example. (4)

(ii) Prove that if a graph $G$ (either connected or not) has exactly 2 vertices of odd degree, then there is a path joining these 2 vertices. (6)

6. (a) (i) Define spanning tree with an example. (4)

(ii) Prove that if $T$ is a tree with $n$ vertices, then it has precisely $n-1$ edges. (6)

(b) Prove that $K \leq K^1 \leq S$. (10)

7. (a) Prove that a graph is bipartite if and only if it contains no odd cycle. (10)

(b) Prove that an edge $e$ of $G$ is a cut edge of $G$ if and only if $e$ is contained in no cycle of $G$. (10)

8. (a) Prove that a non empty connected graph is Eulerian if and only if it has no vertices of odd degree. (10)

(b) (i) Explain max-flow min-cut theorem. (7)

(ii) State Euler's formula. (3)

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